Federal Renewal Compliance Enforcement and Medicaid Coverage Retention During the 2023 Unwinding

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Structured Abstract

Objective

To estimate whether Centers for Medicare and Medicaid Services (CMS)-concurred pauses in procedural disenrollments during the 2023 Medicaid unwinding were associated with state renewal outcomes.

Study Setting and Design

We conducted an observational state-month study covering 51 jurisdictions from February 2023 through July 2024. The primary design was difference-in-differences with state and month fixed effects, exploiting staggered timing of CMS-concurred pause periods across 15 states. We compared these as-treated estimates with intent-to-treat estimates based on September 2023 CMS noncompliance classifications and with two-stage staggered-treatment estimates.

Data Sources and Analytic Sample

We combined publicly available CMS unwinding renewal metrics, CMS enrollment performance indicators, the September 21, 2023 CMS state compliance assessment, the CMS pause-delay chart, and 2022 American Community Survey controls. The balanced state-month skeleton contained 918 observations; the main regressions used 764 state-months with nonmissing renewal outcomes.

Principal Findings

In as-treated models, pause states had a 6.5-percentage-point higher renewal completion rate than non-pause states (95% CI, 0.2 to 12.8 percentage points). Point estimates were directionally favorable but imprecise for procedural termination rates (-5.6 percentage points; 95% CI, -13.0 to 1.9) and total disenrollment rates (-4.4 percentage points; 95% CI, -9.7 to 0.9). The intent-to-treat estimate for renewal completion was essentially null (0.0 percentage points; 95% CI, -6.5 to 6.4). Formal pretrend tests rejected for renewal completion and total disenrollment in the baseline event-study window. The renewal-completion estimate did not survive Benjamini-Hochberg adjustment (adjusted p = 0.52), and the observed effect size was below the approximate 80 percent minimum detectable effect of 8.1 percentage points.

Conclusions

CMS-concurred pauses were associated with higher renewal completion in pause states, but the state-level evidence is sensitive to pretrend, multiple-testing, and power diagnostics. The study is most useful as a quantitative benchmark for a rare federal enforcement episode rather than as strong causal evidence about federal oversight effects.

Keywords: Medicaid; eligibility redetermination; unwinding; administrative burden; federalism; coverage retention; procedural disenrollment

Callout Box

What is known on this topic

- Most Medicaid unwinding disenrollments were procedural rather than eligibility-based.
- Prior unwinding studies focused on coverage loss or automation assistance, not direct federal compliance enforcement.
- CMS-concurred pauses were a rare operational response to documented renewal-processing noncompliance.

What this study adds

- Pause states had higher renewal completion, but broader CMS noncompliance status alone showed no detectable average effect.
- The headline association failed baseline pretrend and multiple-testing diagnostics and was underpowered for modest effects.
- Better beneficiary-level oversight data are needed to evaluate federal enforcement actions credibly.

Introduction

The resumption of Medicaid eligibility redeterminations in 2023 produced one of the largest coverage disruptions in the program's history. More than 25 million people were disenrolled during the unwinding, and roughly 69 percent of terminations were reported as procedural rather than based on an ineligibility determination.[1,2] That pattern renewed attention to the administrative burden embedded in Medicaid renewal processes and to the role of federal oversight when state renewal systems malfunction.[3,4]

In August 2023, CMS identified a specific compliance problem: some states were conducting automatic renewals at the household level rather than the federally required individual level.[5] Under household-level processing, an ineligible or non-responding family member could trigger procedural disenrollment for another member who remained individually eligible. CMS required states to assess their systems, publicly released a compliance table in September 2023, and concurred in pause or delay periods for 15 states while corrective action proceeded.[6,7] By early 2024, federal and state officials reported that more than 400,000 beneficiaries had been reinstated after inappropriate terminations tied to this issue.[7] Later in 2023, CMS also formalized broader renewal-enforcement authorities through interim final rulemaking.[17]

This episode is important because it sits at the intersection of two literatures that have rarely been joined empirically. The administrative burden literature shows that renewal requirements and documentation processes can meaningfully reduce program participation, especially among populations with limited time, information, or administrative support.[3,4,8,9] The Medicaid federalism literature, by contrast, has mostly described federal oversight qualitatively, emphasizing cooperative engagement and the political limits of aggressive enforcement in a joint federal-state program.[10] What has been missing is a quantitative assessment of a concrete federal compliance action that directly altered the procedural termination pipeline during a major coverage transition.

The unwinding period offers a plausible setting for that assessment because the compliance intervention was public, time-bounded, and accompanied by observable state-month outcome measures. At the same time, this is not a clean natural experiment. States with pause periods differed from states without pauses, several pauses began before the September 2023 compliance snapshot, and broader noncompliance status did not cleanly line up with pause adoption. The relevant question is therefore not whether the episode delivers definitive causal identification, but whether careful state-month comparisons can clarify the size, direction, and fragility of the observed associations.

We address three questions. First, were CMS-concurred pauses associated with lower procedural termination and disenrollment rates or higher renewal completion rates? Second, did the broader CMS noncompliance designation have detectable average effects apart from the pause action itself? Third, how much do pretrend, sensitivity, and power diagnostics change the interpretation of any apparent treatment effect? These questions matter both for unwinding policy and for the broader design of federal oversight tools in Medicaid.

Methods

Study Design and Data Sources

We constructed a state-month panel for all 50 states and the District of Columbia covering February 2023 through July 2024. The analysis used five public data sources. First, CMS unwinding renewal metrics from data.medicaid.gov provided monthly counts of renewals due, renewals completed, ex parte renewals, procedural disenrollments, ineligibility disenrollments, and pending renewals.[5] Second, CMS enrollment performance indicators provided monthly enrollment totals and operational measures. Third, the September 21, 2023 CMS state assessment table identified jurisdictions that were noncompliant or still assessing their individual-level renewal processes.[6] Fourth, the CMS pause-delay chart documented the states that obtained CMS concurrence to pause or delay procedural disenrollments and the timing and stated scope of those actions.[7] Fifth, we merged 2022 American Community Survey state characteristics for descriptive balance assessment and selected sensitivities.

The state-month skeleton contained 918 observations. Outcome reporting was structurally missing in some early months, largely because not all states had yet initiated unwinding renewals. We did not impute those values. The main regression sample therefore included 764 state-month observations with nonmissing renewal outcomes. Because the study used publicly available, aggregated administrative data and did not involve human participants, institutional review board review was not required.

Exposure Definitions

We used two complementary treatment definitions. The intent-to-treat specification classified 30 jurisdictions as treated if CMS identified them as noncompliant or still assessing in September 2023 and interacted that indicator with post-September 2023 months. This specification estimates whether the broader compliance designation translated into detectable outcome changes.

The as-treated specification focused on the operational response that is most closely tied to the enforcement episode: CMS-concurred pauses or delays in procedural disenrollments. A state was coded as treated beginning in the first month of its pause period and remained treated while the documented pause was active. Fifteen states entered treatment under this definition, with start dates ranging from May 2023 through October 2023. Because 10 pause states began before the September 2023 assessment snapshot, we prespecified a sensitivity analysis that excluded those early-pause states.

Outcomes

The five primary outcomes were measured as shares using renewals due as the denominator when appropriate: procedural termination rate, total disenrollment rate, renewal completion rate, ex parte renewal rate, and pending renewal rate at month end. These outcomes capture different points in the renewal process. Procedural termination and disenrollment rates reflect coverage loss, renewal completion reflects successful case processing, ex parte renewal reflects automated processing, and pending renewals capture administrative backlog.

Analytic Approach

The main analysis used two-way fixed-effects difference-in-differences models with state and month fixed effects and state-clustered standard errors.[11] We estimated each outcome first under the intent-to-treat definition and then under the as-treated pause definition. Because treatment timing varied across states, we also implemented Gardner's two-stage estimator as a sensitivity check on the as-treated specification.[12]

We used event-study models to assess pretrends and dynamic treatment effects for the as-treated specification. The baseline event window covered six leads and 12 lags relative to pause onset, with the month before onset as the reference period. To clarify whether the main findings depended on a single early lead coefficient, we also re-estimated event studies in a trimmed window of four leads and 12 lags.

Robustness and Diagnostic Analyses

The paper's interpretation depends heavily on diagnostics, so we treated them as part of the main design rather than as an afterthought. We conducted formal pretrend tests, 500 placebo reassignments of pseudo-pause dates, Rambachan-Roth sensitivity analysis for violations of parallel trends, and Oster bounds for sensitivity to unobserved confounding.[13],[14] We also estimated a balanced-panel restriction, an early-pause exclusion, and multiple-testing corrections across the 15 main treatment-effect estimates. To assess whether the design was capable of detecting modest effects, we computed an approximate 80 percent minimum detectable effect for the as-treated design using the outcome standard deviation, treated-state count, and state clustering structure.

Finally, because the intervention was a temporary pause rather than permanent treatment adoption, we went beyond a single binary indicator. A spell decomposition separated the first active month of a pause, subsequent active pause months, and post-pause months. An exploratory intensity specification allowed the active-pause association to vary with planned pause duration and with whether the pause was partial in scope or targeted to a subset of beneficiaries. These analyses were intended to clarify mechanism and timing, not to rescue the design if the main assumptions failed. We did not use Sun-Abraham or Callaway-Sant'Anna as primary estimators because those frameworks are most naturally aligned with absorbing treatment adoption, whereas pause exposure in this setting turns on and off within states.[18],[19]

Results

Descriptive Findings

Descriptive differences between treated and untreated jurisdictions already suggest caution (Table 1). Before September 2023, the broader CMS noncompliance group had lower procedural termination rates and higher renewal completion rates than compliant states, and it also differed meaningfully on baseline state characteristics. For example, treated states had lower poverty rates and higher median household incomes than comparison states, with normalized differences of 0.91 and 0.85 in absolute value. Those differences do not invalidate a fixed-effects design, but they do argue against interpreting pause adoption or noncompliance status as quasi-random.

The treatment definitions also only partially overlapped. Of the 15 pause states, 10 were in the CMS noncompliance or still-assessing group and 5 were classified as compliant. Conversely, 20 ITT-treated states never adopted a documented pause. That mismatch is central to interpreting the null intent-to-treat results below.

Figure 1 plots raw cohort-specific outcome trends over calendar time. The control states and the pause cohorts tracked each other closely before the federal enforcement event, with visible separation on renewal completion appearing only after the cohort-specific pause onset. The figure also shows that late-period patterns are carried by only a handful of cohorts, which motivates caution in over-interpreting the dynamic coefficients. Figure 2 makes the cohort support explicit: pause starts are concentrated in May, June, August, and September 2023, and event-time support at k = 10-12 is driven by the three earliest-onset states. Readers should keep that thin late-period support in mind when interpreting the event-study plots that follow.

Main Treatment Estimates

Table 2 reports the main outcome models. In the as-treated specification, pause states had a 6.5-percentage-point higher renewal completion rate than non-pause states (95 percent confidence interval, 0.2 to 12.8 percentage points). The corresponding Gardner two-stage estimate was 6.4 percentage points and slightly less precise. By contrast, the intent-to-treat estimate for renewal completion was essentially zero.

For the two outcomes most closely tied to coverage loss, the signs were favorable but the estimates were imprecise. The as-treated procedural termination estimate was -5.6 percentage points (95 percent confidence interval, -13.0 to 1.9), and the total disenrollment estimate was -4.4 percentage points (95 percent confidence interval, -9.7 to 0.9). Ex parte renewal rates were near zero, which is substantively useful: the enforcement episode appears more closely related to slowing or interrupting procedural terminations than to increasing automated renewal processing. Pending renewal rates were slightly lower in pause states, but those estimates were also imprecise.

The intent-to-treat results help clarify what the as-treated coefficient can and cannot mean. If the September 2023 compliance designation itself had triggered broad system-wide improvements, the ITT estimates should have shown at least attenuated movement in the same direction as the pause models. Instead, the ITT specification was effectively null for every outcome. A decomposition of the treated states showed that the 20 non-pause jurisdictions in the ITT-treated group were flat on renewal completion relative to compliant non-pause states, while the positive association was concentrated in the subset that actually paused procedural disenrollments.

Event Study and Sensitivity Evidence

The event-study plots are informative but not fully reassuring. Figure 3 shows the renewal-completion dynamic coefficients; the large, negative k = -5 lead is visible and is the main driver of the formal pretrend rejection (Wald p = 0.01). Renewal completion and total disenrollment failed the formal pretrend test in the baseline event window, whereas the other three outcomes did not. For renewal completion, the rejection was driven largely by the earliest lead. Trimming the event window from six to four leads raised the pretrend p-value from 0.01 to 0.25, but the first post-pause coefficient remained small and imprecise. That pattern suggests the pretrend problem may be concentrated rather than pervasive, but it does not resolve the identifying concern.

Other diagnostics point in the same direction. Placebo tests were supportive for renewal completion, with the observed estimate exceeding 96 percent of placebo estimates, yet Rambachan-Roth confidence intervals already included zero under exact parallel trends and widened quickly under modest relaxations of that assumption (Figure 4). The renewal-completion interval at the calibrated M = 1 bound spanned -24.0 to 34.5 percentage points, covering the full range of the point estimate and well beyond. Oster's delta-star for renewal completion was 0.033, implying that unobserved confounders with roughly three percent of the explanatory power of the observed covariates could fully explain away the point estimate. Taken together with the pretrend and multiple-testing results, these sensitivity checks substantially weaken any causal reading of the headline association.

The multiple-testing and power diagnostics further temper the headline result. After Benjamini-Hochberg adjustment across the 15 main tests, the renewal-completion association had an adjusted p-value of 0.52; under Bonferroni correction it was 0.75. The approximate 80 percent minimum detectable effect for renewal completion was 8.1 percentage points, above the observed 6.5-point estimate. On this design, a meaningful effect may exist and still be hard to distinguish from noise, but the reverse is also true: a borderline nominal result should not be treated as conclusive.

Additional Design Checks

Several additional checks help characterize the intervention rather than overturn the main message. Restricting the sample to a balanced reporting panel attenuated the renewal-completion association to 4.9 percentage points. Excluding the 10 pre-September pause states left only 5 treated states; the renewal-completion estimate stayed positive at 7.4 percentage points but became more uncertain. Those results weaken the argument that early pause timing alone explains the main finding, but they also show how dependent the design is on a small number of treated jurisdictions.

The spell decomposition was more informative about mechanism than about identification. Renewal completion rose sharply in the first active month of a pause and remained positive during later active-pause months. Post-pause coefficients for procedural termination and total disenrollment remained negative and imprecise rather than showing an obvious catch-up spike. That pattern is more consistent with at least some prevented coverage loss than with a purely mechanical delay, although post-pause leverage was limited because many states remained paused through the end of the observed period.

The exploratory intensity analysis suggests that planned pause duration mattered more than pause label alone. Longer planned pauses were associated with larger reductions in procedural terminations and total disenrollment, whereas narrower partial-scope or targeted-population pauses looked less favorable on procedural terminations. Because scope and targeted population were not separately identified in the available panel, we interpret those estimates as hypothesis-generating rather than as policy-ready causal effects.

Discussion

This study offers a quantitative benchmark for a rare federal enforcement episode during the Medicaid unwinding. The most policy-relevant positive signal is the as-treated renewal-completion estimate: pause states completed more renewals while pauses were active. That effect size is similar in magnitude to prior evidence on broader unwinding coverage change and operational renewal interventions, but the evidentiary footing is weaker here because the null intent-to-treat result, pretrend rejection, multiple-testing adjustment, and limited power all narrow what the estimate can support.[15],[16]

The comparison with related work is instructive. Herd and colleagues studied a collaborative federal technical-assistance effort aimed at improving automated renewals and found larger and cleaner effects on ex parte processing and procedural denials.[16] Myerson and colleagues showed experimentally that individualized navigation support can improve renewal outcomes for beneficiaries facing administrative barriers.[8] Our study speaks to a different margin: whether a federal enforcement response that slows the disenrollment pipeline is associated with better renewal outcomes at the state level. The answer is plausibly yes for the pause states, but not in a way that justifies strong causal claims about federal compliance oversight in general.

Three substantive implications follow. First, the action that seems to matter is not being labeled noncompliant but actually implementing an operational pause. The null intent-to-treat result suggests that federal designation alone was not enough to shift average outcomes. Second, the direction of the estimates remains consistent with the administrative-burden framework: giving states and beneficiaries more processing time appears more relevant than changing ex parte automation rates in this episode. Third, the results illustrate how difficult it is to evaluate oversight using only aggregated state-month data. Without beneficiary-level information on who was affected, paused, reinstated, and later disenrolled, even a nationally visible intervention remains only partially observable.

The study also has important limitations. The design relies on only 15 pause states, and the main outcome models are underpowered for modest effects. Balance differences, baseline pretrend failures, and small Oster bounds weaken the case for a strong quasi-experimental interpretation. The treatment itself is heterogeneous and temporary: states differed in pause timing, duration, scope, and the populations affected, while other unwinding mitigation efforts continued in parallel. Finally, the state-month outcomes cannot distinguish whether pauses prevented inappropriate disenrollments, accelerated reinstatements, or merely delayed coverage loss.

Those limitations point to a specific policy recommendation rather than a generic request for more research. If CMS intends to use corrective-action pauses or analogous enforcement tools in future renewal crises, it should pair them with standardized beneficiary-level reporting on affected caseloads, pause start and end dates, reinstatements, and post-pause dispositions. That would allow federal and state officials to evaluate whether enforcement altered eligibility processing, slowed inappropriate disenrollment, or simply deferred it. Until those data exist, the present study should be read as disciplined but ultimately suggestive evidence on a high-salience oversight episode.

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Tables and Figures

Table 1. Pre-September 2023 descriptive differences by CMS non-compliance status

Variable	Treated mean	Control mean	Normalized difference
Procedural termination rate	0.655	0.742	-0.46

Variable	Treated mean	Control mean	Normalized difference
Total disenrollment rate	0.351	0.399	-0.33
Renewal completion rate	0.593	0.555	0.25
Ex parte renewal rate	0.476	0.549	-0.30
Pending renewal rate	0.056	0.046	0.13
Poverty rate (%)	11.2	13.3	-0.91
Median household income ($)	78,388	68,813	0.85
Baseline enrollment (millions)	1.43	2.43	-0.42

Notes: Treated jurisdictions are the 30 states classified as noncompliant or still assessing in the September 21, 2023 CMS assessment. Control jurisdictions are the 21 states attesting to compliance. Normalized differences are the difference in means divided by the square root of the average group variance.

Table 2. Main difference-in-differences estimates

Outcome	ITT TWFE coefficient	As-treated TWFE coefficient	DID2S coefficient
Procedural termination rate	0.037	-0.056	-0.031
Total disenrollment rate	0.012	-0.044	-0.027
Renewal completion rate	-0.000	0.065	0.064
Ex parte renewal rate	0.028	0.005	0.008
Pending renewal rate	-0.011	-0.021	-0.038

Notes: All models include state and month fixed effects with standard errors clustered at the state level. ITT treatment equals noncompliance status interacted with post-September 2023 months. As-treated treatment equals an active CMS-concurred pause. DID2S denotes Gardner's two-stage estimator. Detailed standard errors and p-values are reported in the supporting information.

Table 3. Key robustness and diagnostic results for as-treated estimates

Diagnostic	Procedural termination	Total disenrollment	Renewal completion
Baseline pretrend p-value	0.06	0.03	0.01
Trimmed-window pretrend p-value	0.47	0.50	0.25
Placebo empirical p-value	0.25	0.15	0.04
Balanced-panel coefficient	-0.033	-0.024	0.049
Early-pause exclusion coefficient	-0.076	-0.018	0.074
Rambachan-Roth CI at M = 1	[-0.523, 0.317]	[-0.353, 0.235]	[-0.238, 0.345]
Benjamini-Hochberg adjusted p-value	0.53	0.53	0.52
Approximate MDE (percentage points)	12.89	8.18	8.05
Oster delta-star	-0.024	-0.040	0.033

Notes: Coefficients are from the as-treated pause specification. The early-pause exclusion drops the 10 states whose pause periods began before September 2023. MDE denotes the approximate minimum detectable effect at 80 percent power.

Figure 1. Raw cohort-specific outcome trends over calendar time

Figure legend: Mean values of each of the five renewal outcomes plotted over calendar month by pause-start cohort and for the no-pause control group. The vertical dashed line marks the August 30, 2023 CMS federal enforcement event. Pause cohorts and controls tracked each other closely before the enforcement event; visible separation on renewal completion emerged after cohort-specific pause onset.

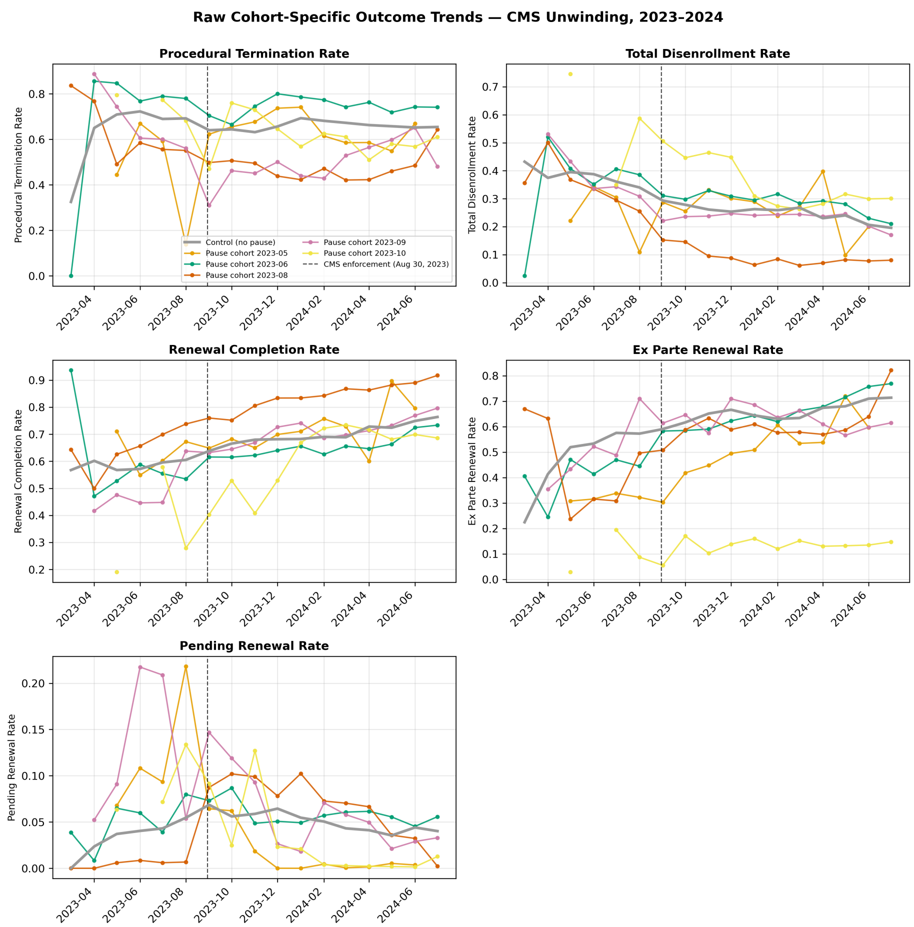

Figure 1: Figure 1. Raw cohort-specific outcome trends

Figure 2. Treatment timing and cohort support

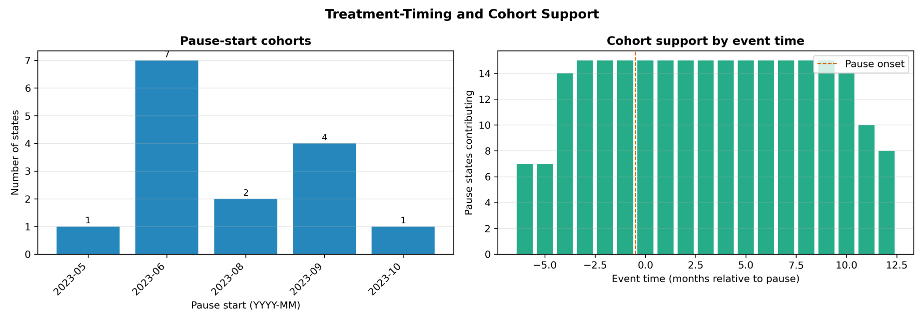

Figure 2: Figure 2. Treatment timing and cohort support

Figure legend: Left panel shows the distribution of CMS-concurred pause start months across the 15 pause states. Right panel shows how many of the 15 pause states contribute an observation at each event time k in the baseline window [-6, 12]. Late-lag coefficients are identified by only a handful of early-onset states.

Figure 3. Event-study estimates for renewal completion around pause onset

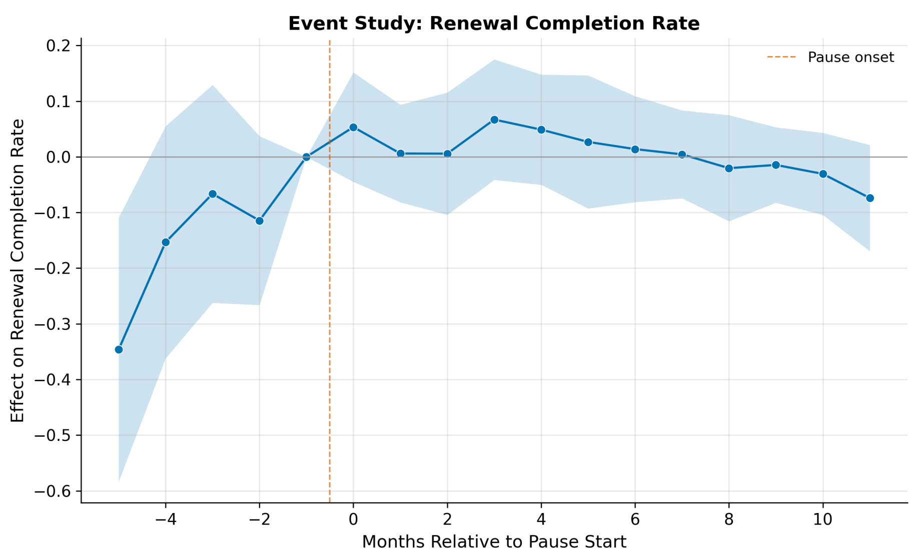

Figure 3: Figure 3. Event-study estimates for renewal completion around pause onset

Figure legend: Points show TWFE event-time coefficients relative to the month before pause onset, with 95 percent pointwise confidence intervals clustered at the state level. The baseline event window uses six leads and 12 lags. The formal Wald pretrend test rejects parallel pretrends (p = 0.01) and the rejection is driven primarily by the k = -5 lead. Bands are pointwise; the dynamic-path interpretation should accordingly be read as descriptive rather than as formal inference on the full path.

Figure 4. Rambachan-Roth sensitivity bounds

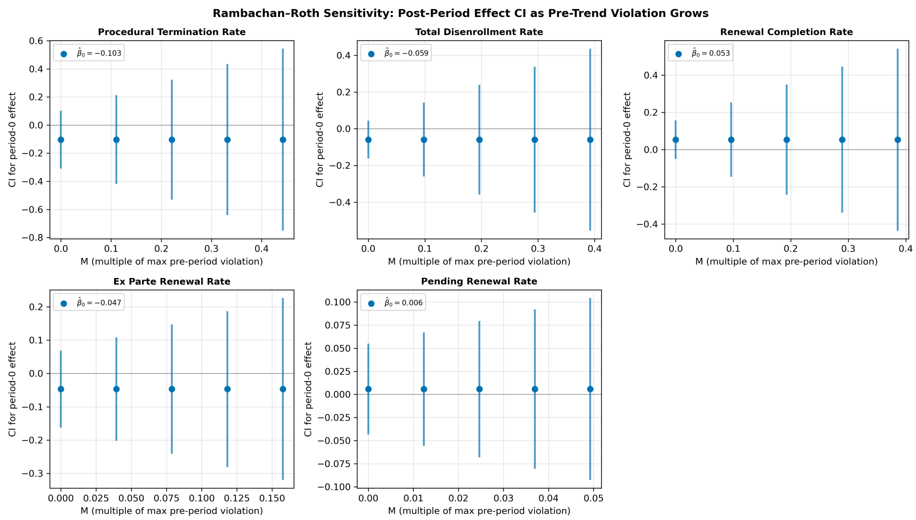

Figure 4: Figure 4. Rambachan-Roth sensitivity bounds

Figure legend: Confidence intervals for the first post-pause coefficient across the five renewal outcomes as the Rambachan-Roth sensitivity parameter M grows from zero (exact parallel trends) through multiples of the maximum observed pre-period slope change. For renewal completion, the CI already spans zero at M = 0 and widens rapidly, underscoring that the headline association is not robust to modest parallel-trends violations.